Modeling **forest fire spread** using machine learning-based cellular automata in a GIS environment

Y Xu, D Li, H Ma, R Lin, F Zhang - Forests, 2022 - mdpi.com

… In this paper, we propose a machine learning-based **forest fire spread model** in which the traditional cellular automata framework is combined with LSSVM (least squares support vector …

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Modeling vorticity-driven **wildfire behavior** using near-field techniques

JJ Sharples, JE Hilton - Frontiers in Mechanical Engineering, 2020 - frontiersin.org

… An example **simulation** using the pyrogenic vector potential coupled to a **wildfire spread simulation** is shown in Figure 2 with s c = 0.5 and s w = 0.5 in Equation (8). These constants …

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Dynamic Risk Assessment of **Wildfire**-Induced Transmission Line Breakdown Based on Data Assimilation Method

Z Wang, M Zha, J Ji, W Wu, L Ding - Fire Technology, 2025 - Springer

… , we will **review** existing related research from two aspects: **wildfire** risk … They calculated **wildfire spread** rates based on the … Besides, various **wildfire propagation** and **behavior simulation** …

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Wildfire spread prediction in north america using satellite imagery and vision transformer

BS Li, R Rad - 2024 IEEE Conference on Artificial Intelligence …, 2024 - ieeexplore.ieee.org

… **Spread** (NDWS) dataset. Originally, NDWS encompassed … , we introduce a **wildfire prediction model** based on a symmet… **review** of operations research methods applicable to **wildfire** …

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Predicting fire **spread** and behaviour on the fireline. **Wildfire** analyst pocket: A mobile app for **wildland fire prediction**

S Monedero, J Ramirez, A Cardil - Ecological Modelling, 2019 - Elsevier

… at providing real-time analysis of **wildfire** behaviour and **spread** in order to support real-… of **spread**, flame length, fireline intensity, crown fire potential, among others) for each **simulation** …

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Describing wildland surface fuel loading for fire **management**: a **review** of approaches, methods and systems

RE Keane - International Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… models are being developed, it is critical that new fuel classification systems be built to balance ecological understanding of fuel **dynamics** with the new input **model** requirements to …

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The large-scale **wildfire spread prediction** using a multi-kernel convolutional neural network

M Marjani, MS Mesgari - ISPRS Annals of the …, 2023 - isprs-annals.copernicus.org

… Lattimer 2019) to predict fire **spread**. To **model** the **spread** of fire, convolutional neural networks … a multi-kernel CNN **model** that can predict **wildfire spread** in large-scale observation data …

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How risk **management** can prevent future **wildfire** disasters in the wildland-urban interface

DE Calkin, JD Cohen, MA Finney… - Proceedings of the …, 2014 - pnas.org

… First, we **review** how the success, or lack thereof, of **wildfire** … fire **behavior** in the nearby wildlands, meaning that proper care of the HIZ separates home losses from **wildland fire behavior**, …

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JP Prestemon, TJ Hawbaker, M Bowden… - Gen. Tech. Rep …, 2013 - research.fs.usda.gov

… a conceptual **model** showing how **wildfire** ignitions are both … variables also influence **wildfire behavior** and **spread**. Thus, … processes affecting **wildfire** ignitions and **behavior** are …

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KP Joshi, S Giri, D Kuinkel, S Kuikel, R Devkota… - Trees, Forests and …, 2025 - Elsevier

… national **forest fire** analyses. This study investigates fire trends, key drivers, and public perceptions of **management** … **Survey** responses highlight strong community support for awareness …

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